

Reflection of my learning experience (Data Science)

The first week of my Data Science Internship was a valuable learning experience that introduced me to the practical use of data to understand a real-world business problem. Through the Employee Attrition Analysis project, I learned that data science involves more than writing code and creating graphs. It requires understanding the problem, exploring data systematically, identifying patterns, and translating findings into meaningful business insights. I gained practical experience using Python, pandas, NumPy, and matplotlib.pyplot in Anaconda and Jupyter Notebook. Using pandas, I loaded the dataset with `pd.read_csv()` and explored its structure using `df.shape`, `df.columns`, and `df.dtypes`. I used `df.isnull().sum()` and `df.duplicated().sum()` to assess data quality, while `df.describe()` and `df.describe(include="str")` helped me understand the numerical and categorical variables. I also used `value_counts()` and `pd.crosstab()` to calculate and compare attrition rates across different employee groups. This taught me the importance of properly inspecting data and using percentages when comparing groups of different sizes. Another important area of learning was data visualization. Using matplotlib.pyplot and pandas plotting functions, I created bar charts, histograms, boxplots, and a pie chart. I learned to use functions such as `plt.title()`, `plt.xlabel()`, `plt.ylabel()`, and `plt.show()` to make visualizations clearer and easier to interpret. I also learned that visualizations should not simply display information but should help answer business questions and support meaningful conclusions. The analysis provided several useful insights. The overall employee attrition rate was 16.12%, while Sales Representatives had a considerably higher attrition rate of 39.76%. Employees who worked overtime had an attrition rate of 30.53%, compared with 10.44% among those who did not. I also observed differences in attrition according to job satisfaction and age. These findings helped me understand how data can reveal patterns that may require further investigation within an organization. Overall, the assignment improved my confidence in using Python and basic data analysis tools while strengthening my analytical and business thinking skills. I now have a better understanding of how raw data can be explored, visualized, and transformed into meaningful insights. Going forward, I would like to strengthen my statistical analysis, data visualization, and predictive modelling skills.

